

Git Hands-on - Exercise 3 (Branching and Merging)

Exercise: Branching and Merging

Aim

To understand Git branching and merging by creating a new branch, making changes in that branch, merging it into the main branch, and deleting the merged branch.

Procedure and Commands

1. Navigate to project folder
`cd /e/DEEPSKILLING/"MODULE 11 GIT"/GitBranchDemo`
2. Initialize repository
`git init`
3. Rename default branch
`git branch -M main`
4. Create README
`touch README.md`
5. Stage README
`git add README.md`
6. Initial commit
`git commit -m "Initial commit"`
7. Create branch
`git branch GitNewBranch`
8. List branches
`git branch`
9. Switch to branch
`git checkout GitNewBranch`
10. Create welcome.txt
`echo "This file is created in GitNewBranch" > welcome.txt`
11. Stage file
`git add welcome.txt`
12. Commit branch changes
`git commit -m "Added welcome.txt in GitNewBranch"`

13. Check status

`git status`

14. Switch to main

`git checkout main`

15. Compare branches

`git diff main GitNewBranch`

16. Merge branch

`git merge GitNewBranch`

17. View history

`git log --oneline --graph --decorate`

18. Delete branch

`git branch -d GitNewBranch`

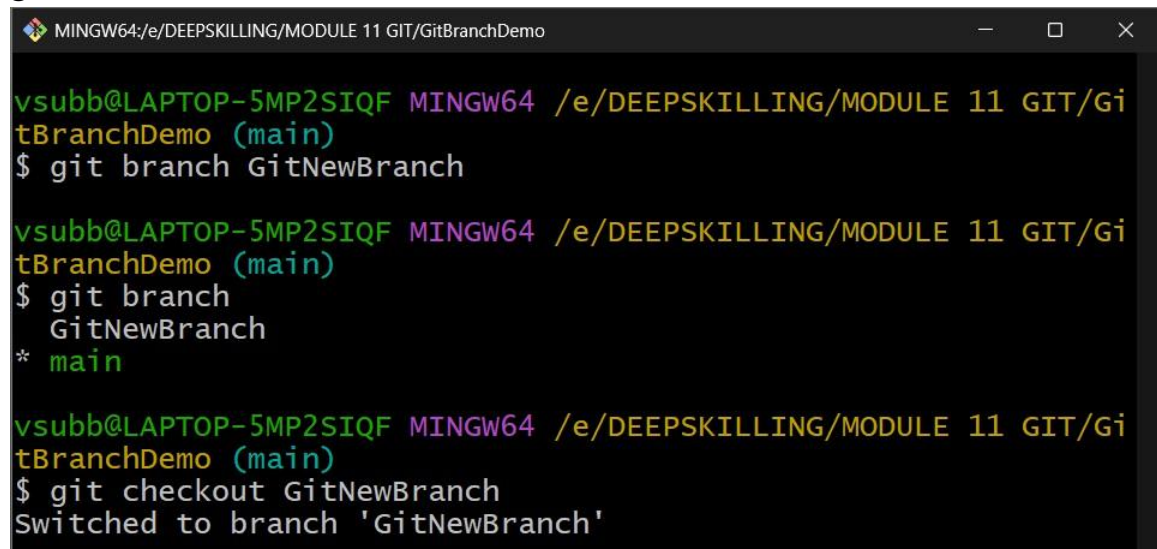
19. Push to GitHub

`git push -u origin main`

Expected Result

The branch is created successfully, changes are committed, merged into the main branch, the commit history shows the merge, and the temporary branch is deleted.

- `git branch`
- `git checkout GitNewBranch`



```
MINGW64:/e/DEEPSKILLING/MODULE 11 GIT/GitBranchDemo
vsubb@LAPTOP-5MP2SIQF MINGW64 /e/DEEPSKILLING/MODULE 11 GIT/GitBranchDemo (main)
$ git branch GitNewBranch

vsubb@LAPTOP-5MP2SIQF MINGW64 /e/DEEPSKILLING/MODULE 11 GIT/GitBranchDemo (main)
$ git branch
  GitNewBranch
* main

vsubb@LAPTOP-5MP2SIQF MINGW64 /e/DEEPSKILLING/MODULE 11 GIT/GitBranchDemo (main)
$ git checkout GitNewBranch
Switched to branch 'GitNewBranch'
```

- git status

```
vsubb@LAPTOP-5MP2SIQF MINGW64 /e/DEEPSKILLING/MODULE 11 GIT/GitNewBranchDemo (GitNewBranch)
$ git status
On branch GitNewBranch
nothing to commit, working tree clean
```

- git diff main GitNewBranch

```
vsubb@LAPTOP-5MP2SIQF MINGW64 /e/DEEPSKILLING/MODULE 11 GIT/GitNewBranchDemo (main)
$ git diff main GitNewBranch
diff --git a/welcome.txt b/welcome.txt
new file mode 100644
index 0000000..319a3b0
--- /dev/null
+++ b/welcome.txt
@@ -0,0 +1 @@
+This file is created in GitNewBranch
```

- git merge GitNewBranch

```
vsubb@LAPTOP-5MP2SIQF MINGW64 /e/DEEPSKILLING/MODULE 11 GIT/GitNewBranchDemo (main)
$ git merge GitNewBranch
Updating 35c866e..aa25439
Fast-forward
 welcome.txt | 1 +
 1 file changed, 1 insertion(+)
 create mode 100644 welcome.txt
```

- git log --oneline --graph --decorate

```
vsubb@LAPTOP-5MP2SIQF MINGW64 /e/DEEPSKILLING/MODULE 11 GIT/GitNewBranchDemo (main)
$ git log --oneline --graph --decorate
* aa25439 (HEAD -> main, GitNewBranch) Added welcome.txt in GitNewBranch
* 35c866e Initial commit
```

- git branch -d GitNewBranch

```
vsubb@LAPTOP-5MP2SIQF MINGW64 /e/DEEPSKILLING/MODULE 11 GIT/GitNewBranchDemo (main)
$ git branch -d GitNewBranch
Deleted branch GitNewBranch (was aa25439).

vsubb@LAPTOP-5MP2SIQF MINGW64 /e/DEEPSKILLING/MODULE 11 GIT/GitNewBranchDemo (main)
$ git branch
* main
```

- GitHub repository after push

Conclusion

The branching and merging operations were completed successfully and the updated repository was pushed to GitHub.